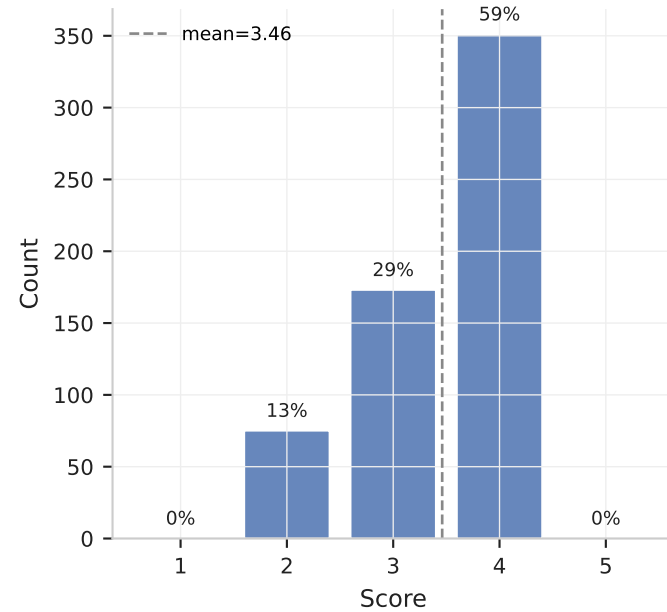
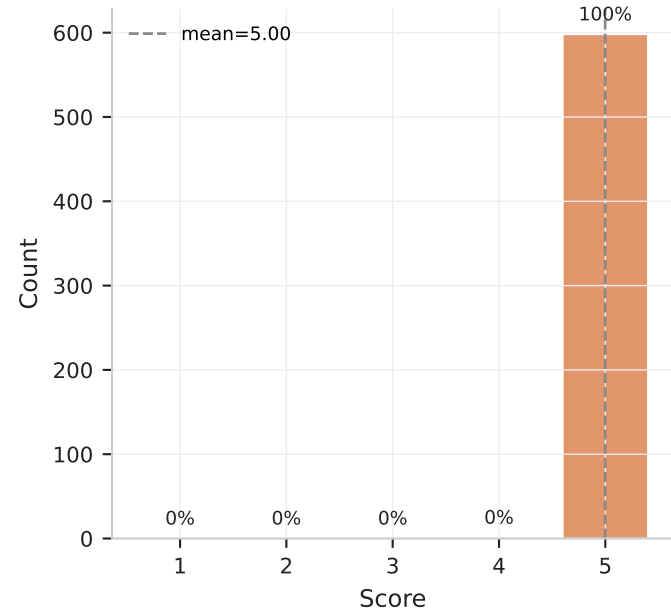


## Score Frequency per Dimension (scores 1-5)

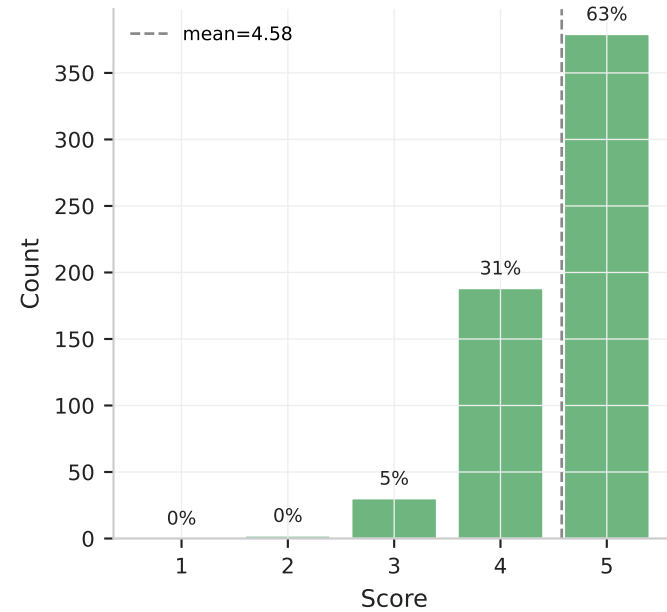
### Socratic Guidance



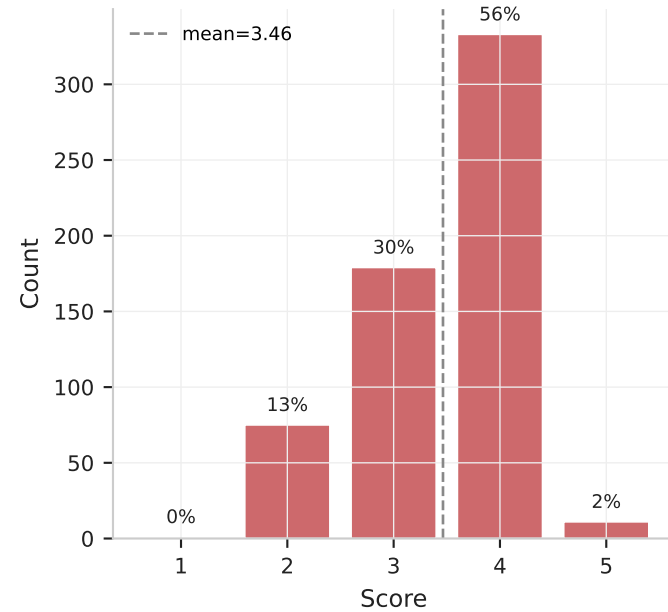
### Mathematical Accuracy



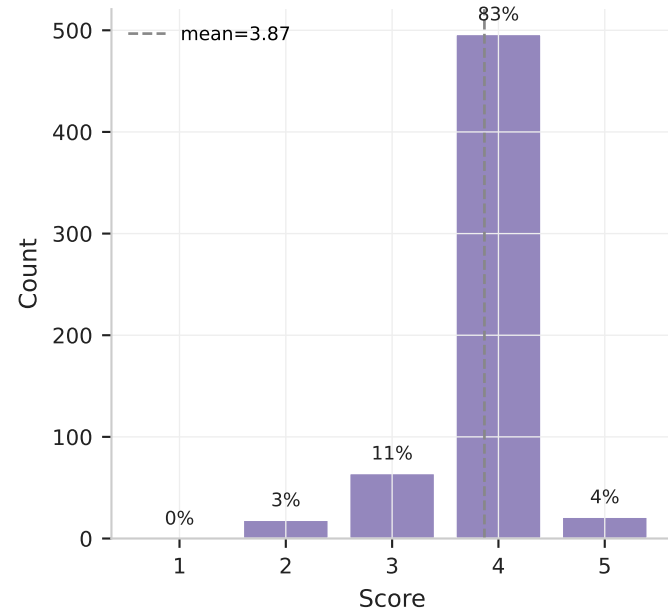
### Relevance



### Conciseness

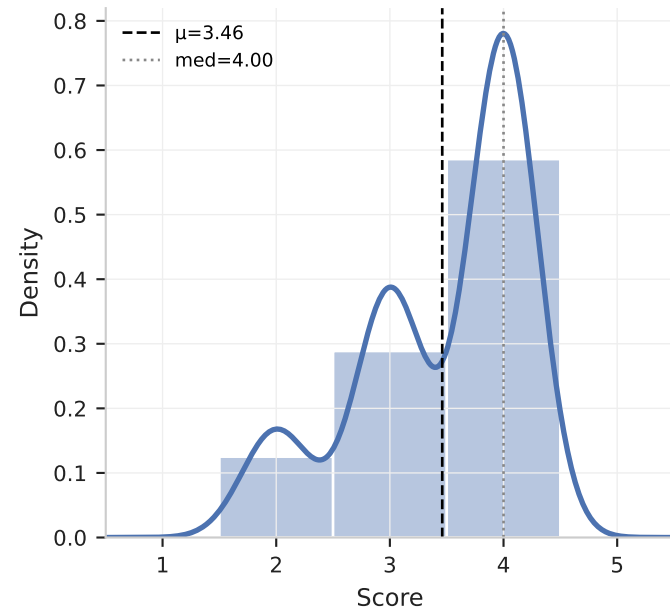


### Overall Quality

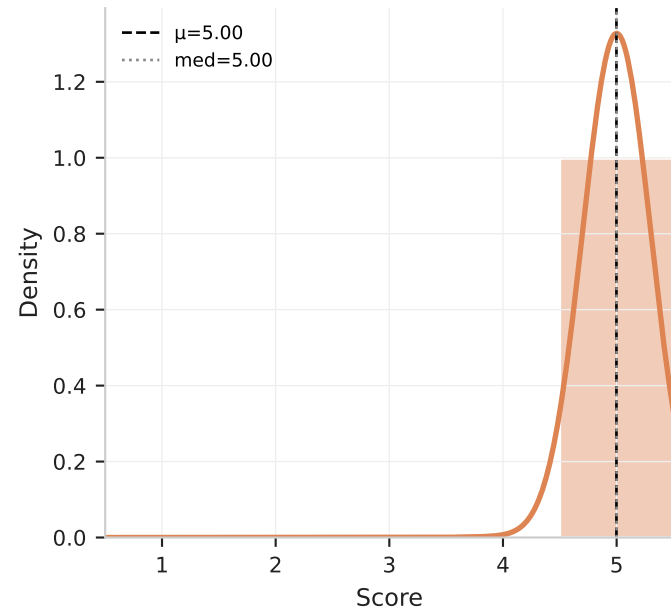


## Score Distribution with KDE (per dimension)

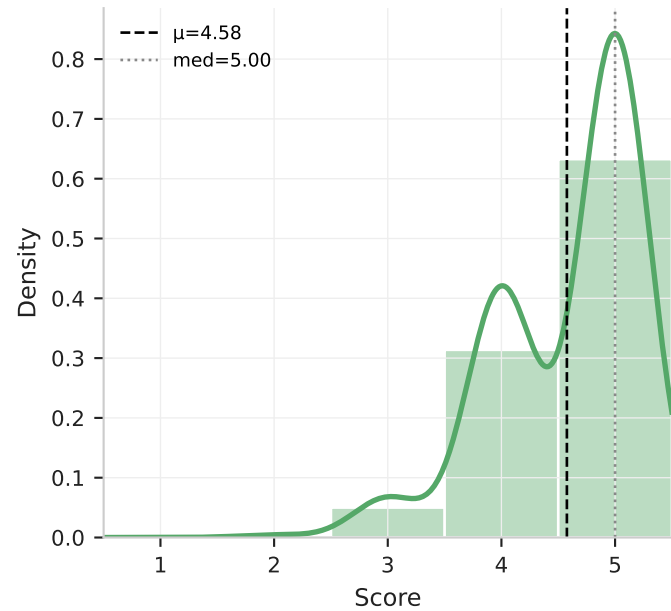
### Socratic Guidance



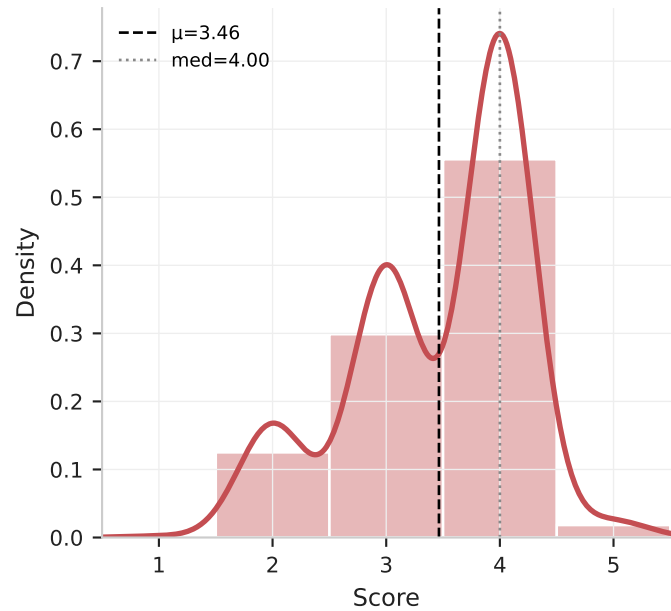
### Mathematical Accuracy



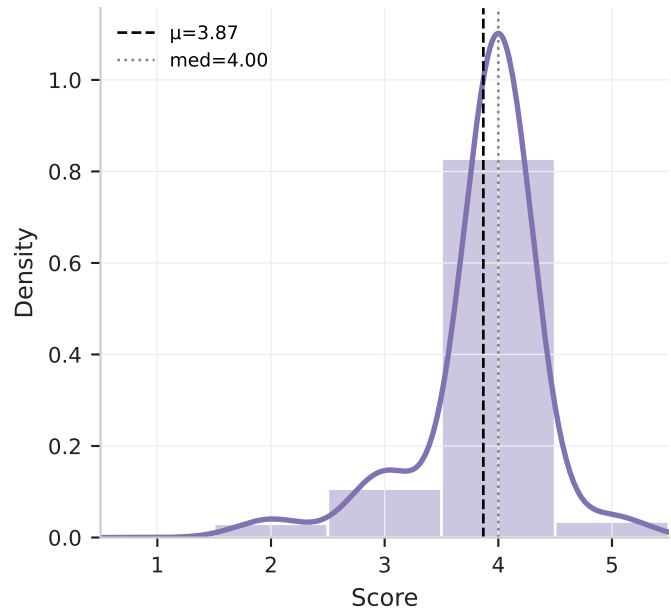
### Relevance



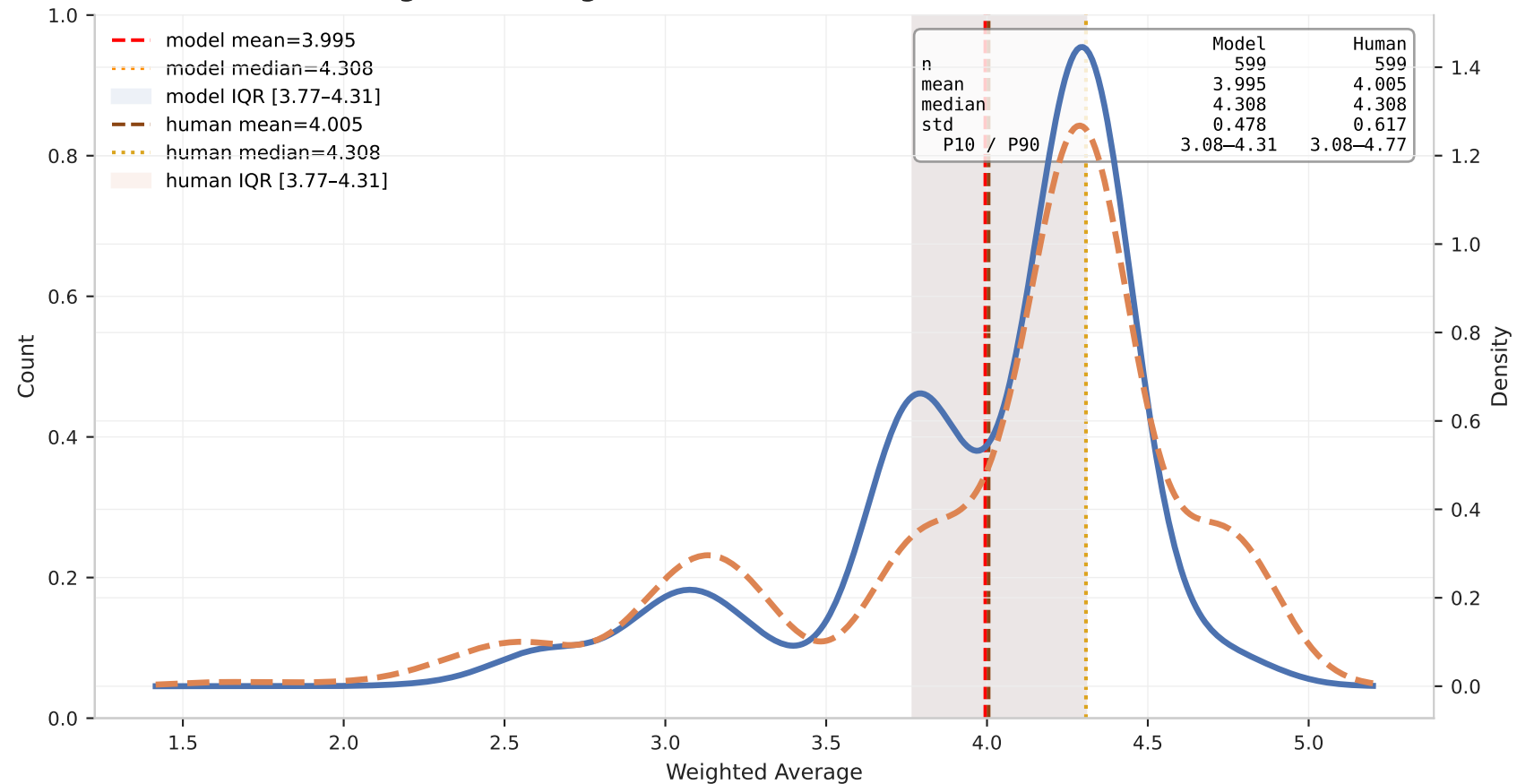
### Conciseness



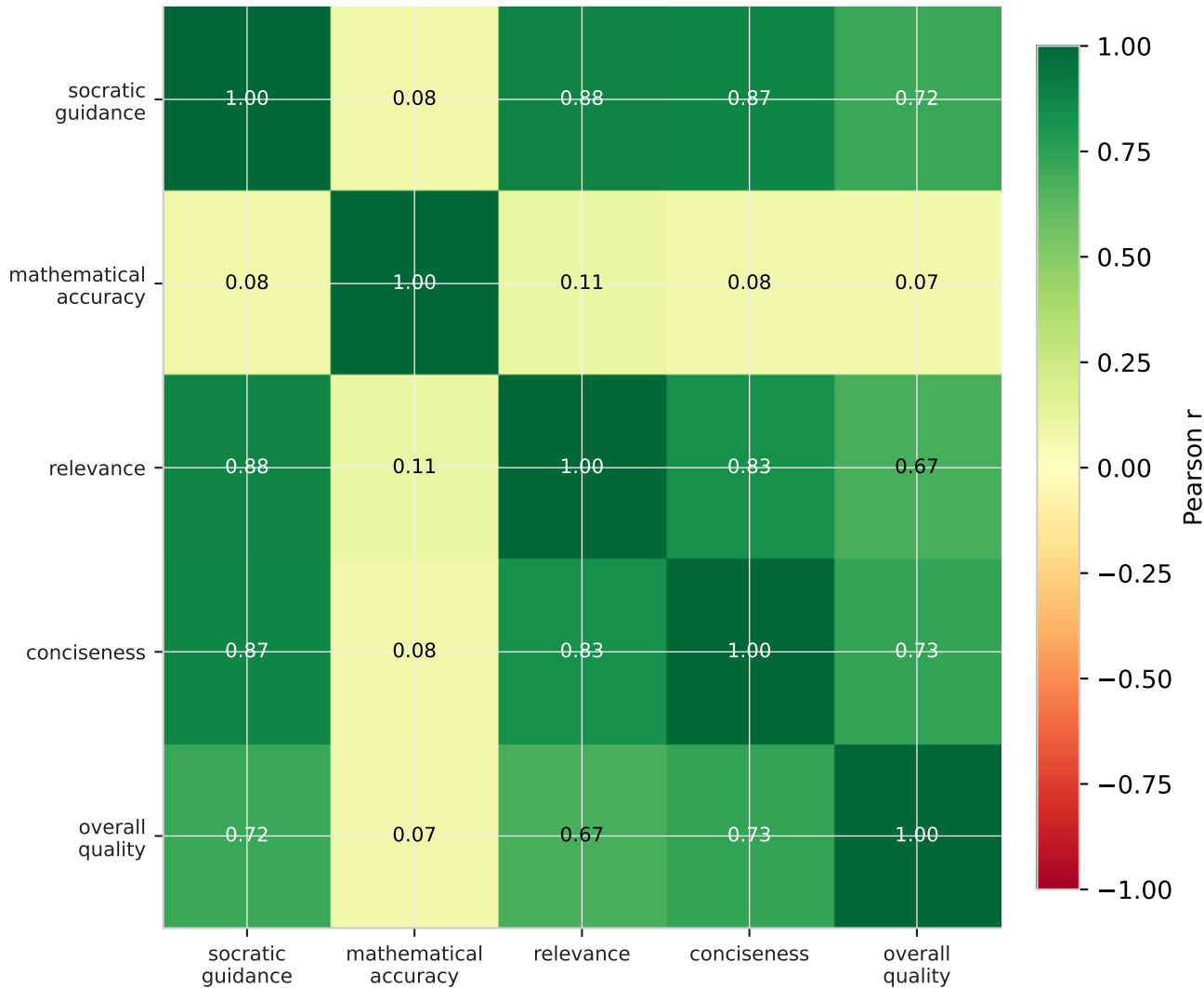
### Overall Quality



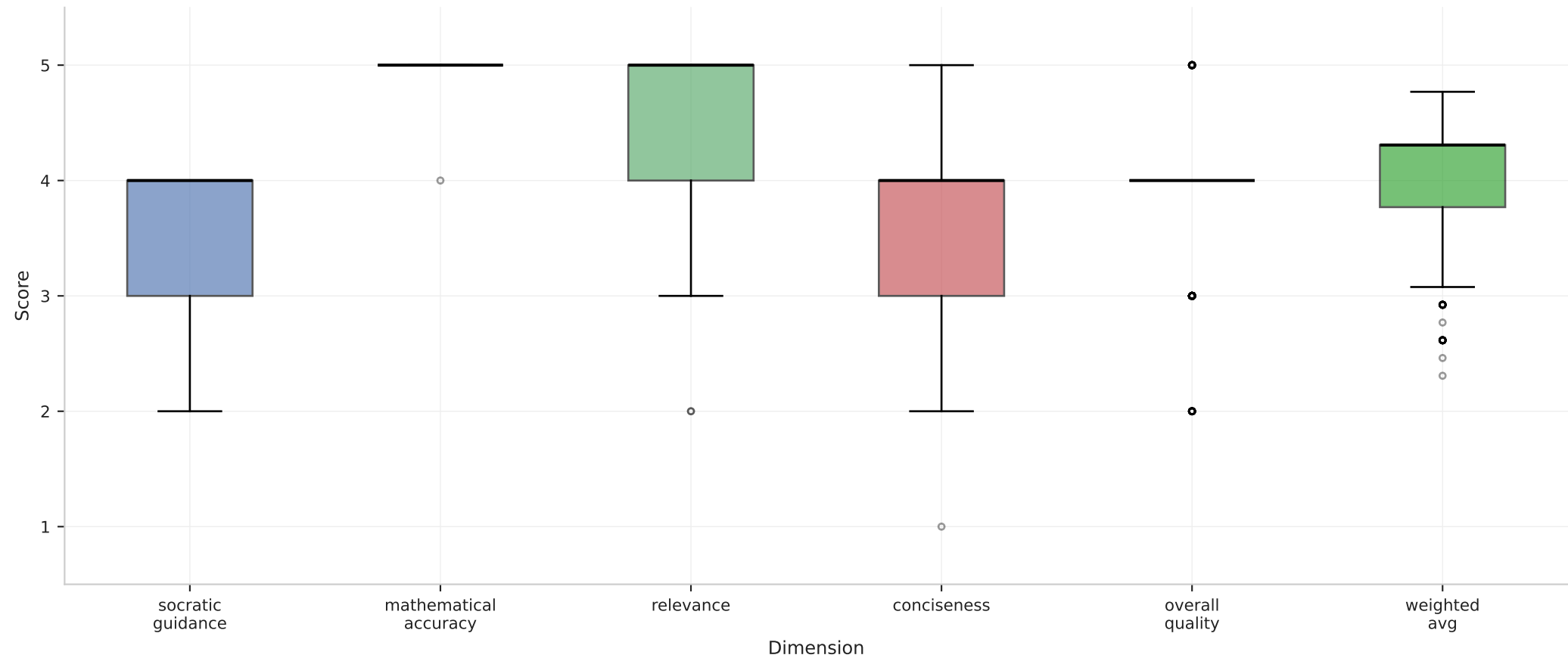
# Weighted Average Score Distribution — Model vs Human



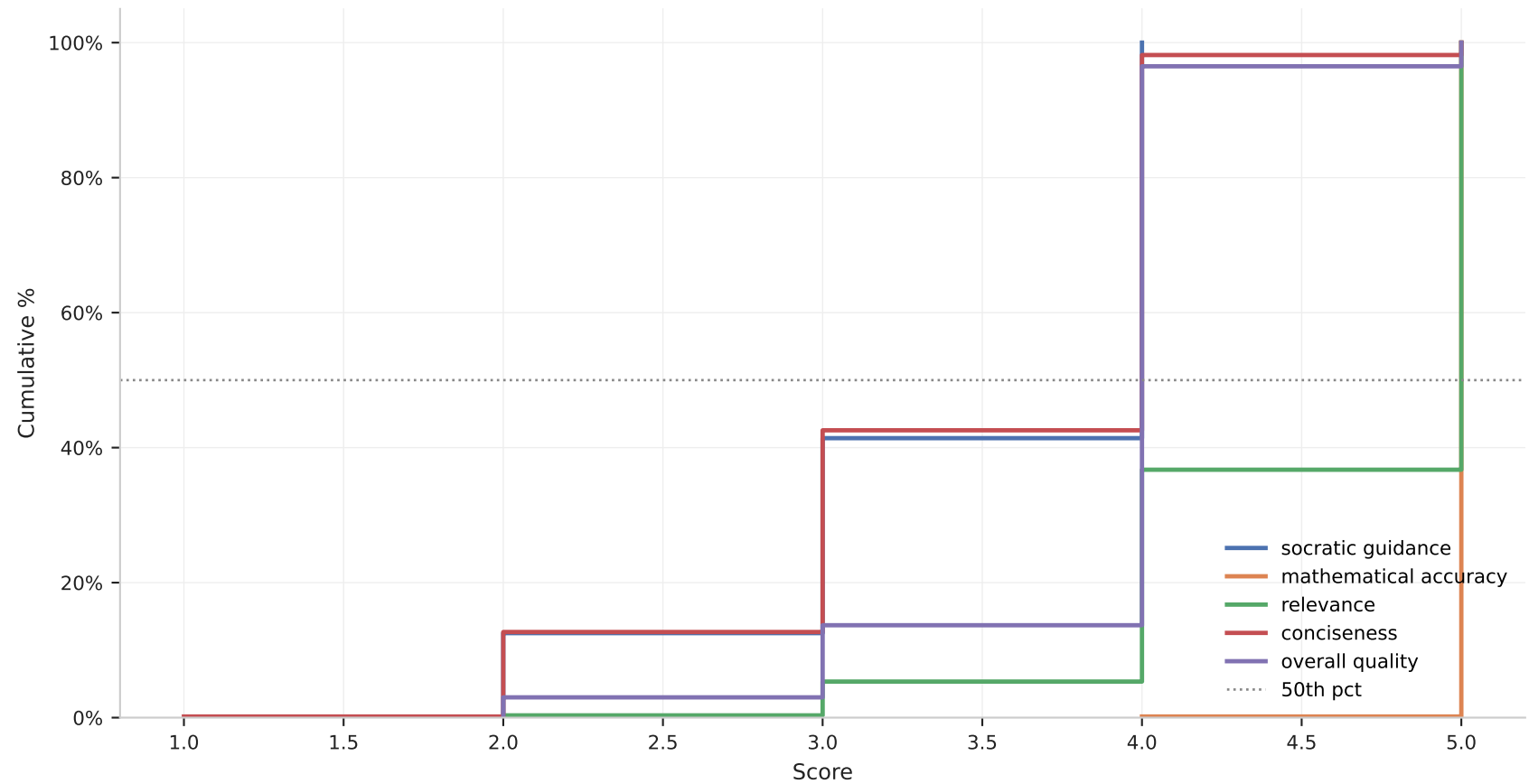
**Cross-Dimension Pearson Correlation Matrix**



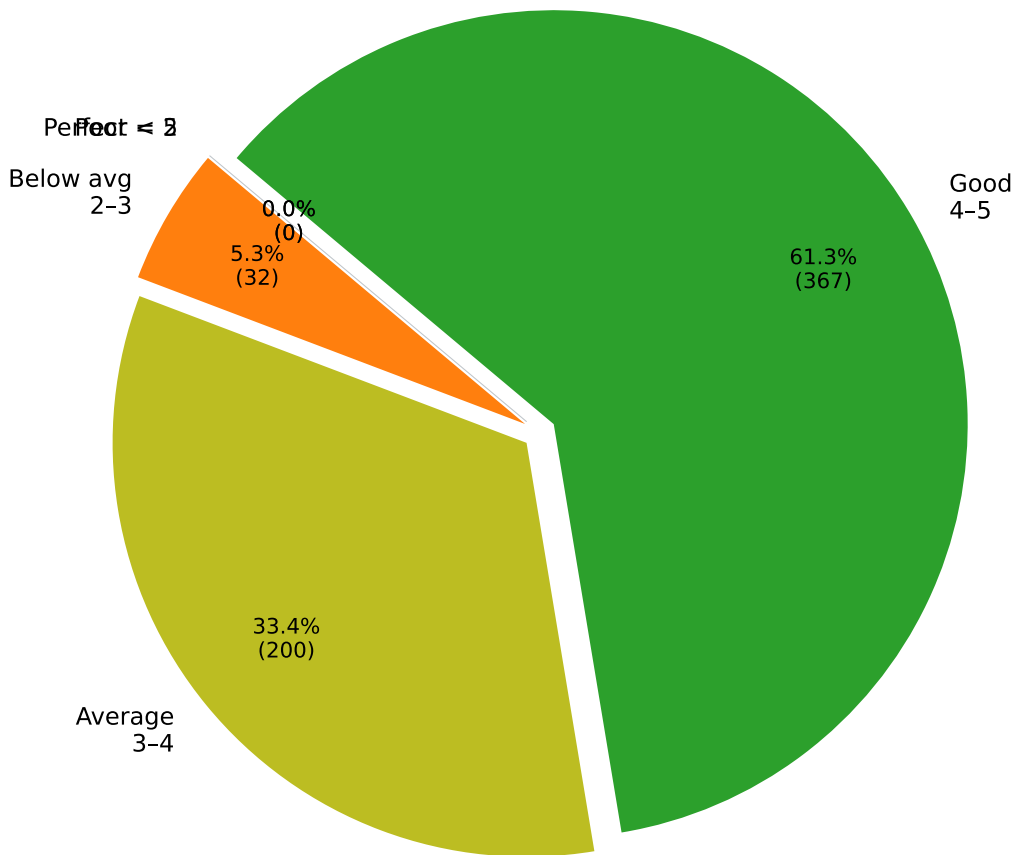
Score Distribution — Box & Whisker (with outliers)



# Empirical Cumulative Distribution Functions



## Grade-Band Distribution (Weighted Average)



Descriptive Statistics Summary

|                       | n   | mean  | median | std   | min  | max  | P25   | P75   |
|-----------------------|-----|-------|--------|-------|------|------|-------|-------|
| socratic guidance     | 599 | 3.461 | 4.000  | 0.706 | 2.00 | 4.00 | 3.000 | 4.000 |
| mathematical accuracy | 599 | 4.998 | 5.000  | 0.041 | 4.00 | 5.00 | 5.000 | 5.000 |
| relevance             | 599 | 4.576 | 5.000  | 0.604 | 2.00 | 5.00 | 4.000 | 5.000 |
| conciseness           | 599 | 3.464 | 4.000  | 0.739 | 1.00 | 5.00 | 3.000 | 4.000 |
| overall quality       | 599 | 3.868 | 4.000  | 0.495 | 2.00 | 5.00 | 4.000 | 4.000 |
| weighted avg          | 599 | 3.995 | 4.308  | 0.478 | 2.31 | 4.77 | 3.769 | 4.308 |